Length as a Primary Confound in AI-vs-Human Stylistic Comparisons: A Conservative Preliminary Report

Abstract

Keywords: stylometry, AI-generated text, computational stylistics, lexical diversity, conjunction frequency, Random Forest classifier, creative writing, multi-model comparison

Do AI-generated text and human-written text differ systematically in structural style, and if so, which differences hold across diverse language models? This preliminary report quantifies stylistic differences using eight locked structural metrics (word length, sentence length, sentence length variance, type-token ratio, MTLD, conjunction frequency, paragraph length, and paragraph length variance), plus five supplementary indicators (lexical density, word-length standard deviation, long-word ratio, sentence-length coefficient of variation, and sentences per paragraph; Section 3.5), applied to 30 creative writing prompts, each paired with one human-authored story and AI-generated outputs from 9 contemporary models, namely GPT-3.5, MiMo-V2.5-Pro, DeepSeek-V4-Flash, MiniMax-M3, Muse-Spark-1.2-Contributor, Kimi-K3, Gemini-3.6-Flash, GPT-5.6-Luna and Claude-Sonnet-5-High, yielding 299 observations. We compare each model's 30 texts against the 30 human texts with independent-samples Welch t-tests (two-tailed; no multiplicity correction; Cohen's d reported for every comparison), quantify within-model homogeneity across the nine core indicators (the eight locked metrics plus MATTR; variance ratios and Levene tests), and train a Random Forest classifier evaluated with StratifiedGroupKFold cross-validation grouped by prompt. AI-generated texts are systematically 1.6× longer than human-written texts in our sample (mean 719 vs 439 words). AI texts also show lower within-model variance than the human group in most metric comparisons — most consistently in the lexical and paragraph layers, such as MATTR, TTR, conjunction frequency and paragraph-length variance (smaller variance in 9/9 models; median variance ratios 0.08–0.56). Length residualization shows that the apparent vocabulary poverty of AI texts is length-attributable: the raw AI-lower TTR pattern (5/9 conditions) does not survive as such — the 4/9 conditions that remain significant after length control reverse to AI > human, aligning TTR with MATTR and MTLD. Conjunction-frequency differences partially persist (5/9 survive pooled residualization; 2/9 under an ANCOVA robustness check). The eight-metric classifier achieves 91.6% raw accuracy but only 68.7% balanced accuracy (MCC = 0.449, AUC = 0.867, human recall 12/30 = 40%), essentially tied with a length-only baseline at 69.4% balanced accuracy. Findings are preliminary

and underpowered for medium effects (30% power at r=0.3); acquisition-channel heterogeneity across four sources, namely direct API, third-party dataset, Manual Arena paste and Faro API, precludes model-level interpretation. Specific items this paper cannot confirm are listed in the Limitations section.

1. Introduction

Large language models have transformed how text is produced across education, journalism, creative writing, and professional communication. As AI-generated text becomes ubiquitous, a natural question arises: does AI writing exhibit systematic structural patterns that distinguish it from human writing? This question matters not only for detection but also for understanding what AI writing "sounds like" at a structural level, independent of content or coherence.

A common observation is that AI-generated text tends to be "patterned" or "formulaic" in its structure, with similar sentence lengths, uniform paragraph structures, and consistent use of transition words. However, these observations have rarely been subjected to rigorous empirical testing across multiple AI systems using controlled, prompt-matched data.

Most existing work on AI text characterization has focused on semantic coherence, factual accuracy, or surface-level detection. Fewer studies have employed computational stylometry, quantitative analysis of writing style features, to systematically compare AI and human text. Stylometric methods have a long history in authorship attribution and forensic linguistics (Juola, 2008), and they offer a principled framework for identifying structural patterns that persist across topics and genres.

This study sits at the intersection of two strands of prior work. First, recent applications of stylometry reach high accuracy in distinguishing human from AI creative writing (O'Sullivan, 2025). Corpus work on prompt-matched stories, in particular the GPT-WritingPrompts collection, documents systematic human-versus-GPT-3.5 differences in character portrayal (Huang et al., 2024); our human baseline and prompt set come from the parent WritingPrompts corpus (Fan et al., 2018), and the shorter stories in the GPT-3.5 condition (531 ± 40 words) reflect that collection's 500-word instruction rather than a model property. Second, studies of LLM creativity find that AI prose is fluent but comparatively lacking in idiosyncratic style (Chakrabarty et al., 2024a) and that writers edit model output to inject personal voice (Chakrabarty et al., 2024b), while genre conventions further shape model output (Lucy & Bamman, 2021); together these findings suggest that AI text may be more uniform than human text, the prediction that our variance-based tests target directly.

A second strand concerns measurement. Vocabulary-richness indices differ in how they treat text length: raw type-token ratio is length-sensitive, which motivated length-controlled alternatives such as MTLD (McCarthy & Jarvis, 2010) and MATTR with a moving window (Covington & McFall, 2010; the window is fixed at 100 tokens in this study). Because length covaries with stylistic variables more broadly, comparisons that ignore it risk reporting the statistics of length rather than of style. Within our hypothesis set, conjunction frequency and vocabulary diversity have the strongest prior support, whereas the evidence on sentence- and paragraph-length variance is more mixed; the report therefore tests four directional hypotheses plus one classification benchmark (H5) across nine contemporary models.

The present study addresses this gap by comparing eight locked structural metrics across 9 contemporary AI models and human-authored stories, all written in response to the same 30 creative writing prompts. This design controls for topic and genre effects through prompt matching, compares each model's outputs against the human baseline as independent groups, and combines hypothesis-driven testing with machine learning classification.

Our study makes three contributions. First, it examines multiple AI systems rather than treating "AI text" as monolithic. Second, it uses a prompt-matched design that controls for prompt-level confounds. Third, it combines independent-samples group tests for specific directional hypotheses with Random Forest classification for overall discriminability.

The analysis focuses on creative short fiction from Reddit's r/WritingPrompts, a genre where structural variation is naturally high and where AI writing assistance is increasingly common. By quantifying which structural stereotypes hold empirically and which do not, this study provides a more nuanced, evidence-based understanding of how AI writing differs from human writing.

2. Method

2.1 Data

The study uses 30 writing prompts from the Reddit WritingPrompts corpus (Fan et al., 2018). For each prompt, we collected one human-authored story and one AI-generated story from each of 9 AI models, yielding up to 30 prompt-matched observations per condition.

Human baseline. We use 30 human-written stories from the WritingPrompts dataset, as processed by Huang et al. (2024), licensed under MIT.

AI conditions. The 9 AI models represent a range of architectures, training approaches, and generation dates:

Model	Provider	Source	N
GPT-3.5	OpenAI	Huang et al. (2024)	30
MiMo-V2.5-Pro	Xiaomi	Command Code Goat	30
DeepSeek-V4-Flash	DeepSeek	Command Code Goat	30
MiniMax-M3	MiniMax	MiniMax CN (Anthropic)	30
Muse-Spark-1.2	Muse	Command Code Goat	30
Kimi-K3	Moonshot	Command Code Goat	29
Gemini-3.6-Flash	Google	Arena + Faro API	30
GPT-5.6-Luna	OpenAI	Manual Arena	30
Claude-Sonnet-5-High	Anthropic	Faro API	30

Kimi-K3 is missing 1 observation (prompt 26, “best friend speaks in rhyme”, repeatedly timed out across multiple API providers). All other AI conditions have complete N=30 coverage.

2.2 Metrics

We extract eight locked structural metrics from each text using spaCy (en_core_web_sm) for tokenization, sentence splitting, and POS tagging, plus custom Python functions for paragraph analysis. Five supplementary indicators at the word, sentence and paragraph levels (Section 3.5) are analysed separately and are not part of the locked set.

#	Metric	Definition	Implementation
1	Word length	Mean characters per word	Average of `len(token.text)` over spaCy tokens
2	Sentence length	Mean words per sentence	Average token count per spaCy sentence
3	Sentence length variance	Population variance of sentence lengths	`numpy.var` over sentence word counts
4	TTR	Type-token ratio	Unique tokens / total tokens
5	MTLD	Measure of Textual Lexical Diversity	Implementation following McCarthy and Jarvis (2010)
6	Conjunction frequency	Conjunctions per word	Count of POS tags CCONJ and SCONJ / total words
7	Paragraph length	Mean words per paragraph	Average word count per paragraph (split on double newline)
8	Paragraph length variance	Population variance of paragraph lengths	`numpy.var` over paragraph word counts

2.3 Hypotheses

We test 4 directional hypotheses using condition-level group comparisons, in which each AI condition's texts are compared as one group against the 30 human texts:

Hypothesis	Metric	Direction	Rationale
H1	Sentence length variance	AI < Human	AI text uses more uniform sentence lengths
H2	Paragraph length variance	AI < Human	AI text uses more uniform paragraph lengths
H3	TTR	AI < Human	AI text is more repetitive in vocabulary
H4	Conjunction frequency	AI > Human	AI text uses more explicit connectives

H5 tests whether a Random Forest classifier using all eight locked metrics achieves balanced accuracy significantly above the 50% chance baseline (permutation test, $p<0.05$, n_perm=1000), evaluated with 5-fold StratifiedGroupKFold cross-validation (group=prompt_id).

2.4 Statistical Analysis

Hypothesis testing. Condition-level comparisons are performed as independent-samples group comparisons: for each AI condition, the model's 30 texts form one group and the 30 human texts form the other, and we apply Welch's independent-samples t-test (two-tailed). No multiple-comparison correction is applied; exact p-values are reported together with Cohen's d for every comparison, and the two should be read jointly — with larger samples, small fluctuations can push p below .05, so the effect size carries the interpretive weight. As a robustness check, every comparison is also run as a Mann-Whitney U (rank-sum) test; across the 81 metric × condition comparisons the two approaches agree at $\alpha = 0.05$ in 72 cases (89%). MATTR is computed with a fixed window of 100; no cross-window comparison is performed, and window-sensitivity checks are reported as a robustness note only.

Classification (H5). We train a Random Forest classifier (scikit-learn, 100 trees, random_state=42, class_weight='balanced') using all eight locked metrics as features. To prevent data leakage from prompt-level grouping, since the same prompt generates both human and AI observations, we evaluate using 5-fold StratifiedGroupKFold cross-validation (group=prompt_id, shuffle=True, random_state=42). We report mean accuracy plus or minus standard deviation across folds, per-condition accuracy, and feature importance rankings. We also report the majority-class baseline, which predicts the majority class for all observations, as a reference point for interpreting classifier accuracy. Statistical significance of the classifier's balanced accuracy is assessed via a group-aware permutation test (n_perm=1000): we shuffle labels within each prompt_id group,

retrain the classifier, and compute the proportion of permuted balanced accuracies that meet or exceed the observed value (one-tailed p-value).

Tools. All analyses use Python 3 with spaCy (Honnibal & Montani, 2017) for tokenization and part-of-speech tagging, pandas, numpy, scipy, and scikit-learn (Pedregosa et al., 2011) for the Random Forest classifier. All analysis code is listed in the Data and Code Availability section.

Software Environment. All experiments were conducted using Python 3.11 with the following package versions: spaCy 3.8.0 (with `en_core_web_sm` 3.8.0), scikit-learn 1.6.0, pandas 2.3.3, numpy 1.26.4, and scipy 1.17.1. Random seeds are set throughout the pipeline: `random.seed(42)` for prompt sampling, and `random_state=42` for the Random Forest classifier and cross-validation splits. This enables full reproducibility of all reported statistics on the same hardware and software environment for the six controlled API conditions; the GPT-3.5, Gemini-3.6-Flash, and GPT-5.6-Luna conditions are snapshot-based and not regenerable from the documentation alone.

Tokenization operationalization. Word length, TTR, and conjunction frequency use only alphabetic tokens (token.is_alpha=True), excluding punctuation, numbers, and whitespace. Word length is the mean character count of alphabetic tokens. TTR is the ratio of unique lowercased alphabetic tokens to total alphabetic tokens, without lemmatization. Conjunction frequency is the count of CCONJ and SCONJ tags among alphabetic tokens divided by total alphabetic tokens. Re-running with this filter yields word length means within ±0.02 characters of the original, for example Human 4.27 versus filtered 4.27, confirming the original analysis is not polluted by punctuation.

MTLD Implementation Note. The MTLD metric was implemented following the algorithm described in McCarthy and Jarvis (2010), using a TTR threshold of 0.72 and segment size of 10. The implementation computes forward and backward pass MTLD and reports their mean. Independent replication using the `lexical_diversity` Python package is recommended for confirmation studies; that package provides the original authors' reference implementation. MATTR Implementation Note. MATTR, the moving-average type–token ratio, follows Covington and McFall (2010), who introduce the moving-window algorithm and show that the resulting measure is independent of text length. The window size is fixed at 100 tokens for every text and every condition; because MATTR values depend on the window size, the window is held constant rather than tuned.

3. Results

3.1 Descriptive Statistics

Tables 1a to 1c present the mean and standard deviation of word count and each stylometric metric across all 9 analyzed AI conditions and the human baseline.

Condition	n	Word count
Claude	30	664 ± 28
DeepSeek	30	688 ± 116
GPT-5.6	30	789 ± 73
Gemini	30	716 ± 65
Kimi	29	674 ± 80
MiniMax	30	778 ± 71
MiMo	30	753 ± 64
Muse	30	878 ± 105
GPT-3.5	30	531 ± 40
Human	30	439 ± 245

Table 1a. Word count (mean ± SD words per text) across all 9 AI conditions and the human baseline.

Condition	Word len.	Sent. len.	Sent. var.	TTR
Claude	4.41 ± 0.33	14.34 ± 3.43	121.2 ± 51.0	0.500 ± 0.04
DeepSeek	4.16 ± 0.26	9.51 ± 1.50	41.0 ± 11.6	0.460 ± 0.06
GPT-5.6	4.52 ± 0.29	7.67 ± 1.67	26.4 ± 8.3	0.450 ± 0.04
Gemini	4.59 ± 0.22	12.71 ± 2.40	54.1 ± 13.3	0.547 ± 0.04
Kimi	4.28 ± 0.21	12.90 ± 2.84	95.6 ± 51.3	0.492 ± 0.03
MiniMax	4.16 ± 0.25	11.89 ± 3.70	81.2 ± 44.0	0.438 ± 0.04
MiMo	4.38 ± 0.23	12.09 ± 3.02	78.9 ± 36.3	0.482 ± 0.03
Muse	4.26 ± 0.24	12.08 ± 3.18	66.2 ± 17.1	0.443 ± 0.04
GPT-3.5	4.91 ± 0.22	13.82 ± 2.17	59.5 ± 21.0	0.504 ± 0.02
Human	4.27 ± 0.31	10.34 ± 3.00	53.5 ± 39.6	0.507 ± 0.08

Table 1b. Descriptive statistics, word- and sentence-level metrics (mean ± SD) across all 9 AI conditions and the human baseline. Word len. = characters/word; Sent. len. = words/sentence; Sent. var. = sentence length variance; TTR = type-token ratio.

Condition	MTLD	Conj. freq.	Para. len.	Para. var.
Claude	14.61 ± 0.39	0.096 ± 0.01	37.4 ± 11.5	597.6 ± 254.5
DeepSeek	14.67 ± 0.38	0.089 ± 0.02	45.4 ± 22.3	657.4 ± 349.0
GPT-5.6	14.36 ± 0.41	0.085 ± 0.02	10.1 ± 9.2	86.5 ± 68.8

Gemini	14.92 ± 0.36	0.080 ± 0.01	38.2 ± 14.6	441.4 ± 188.7
Kimi	14.52 ± 0.48	0.092 ± 0.02	45.3 ± 17.2	780.9 ± 368.7
MiniMax	15.25 ± 0.44	0.094 ± 0.01	38.1 ± 17.9	703.7 ± 362.4
MiMo	14.47 ± 0.42	0.092 ± 0.01	43.6 ± 17.4	656.5 ± 366.1
Muse	14.36 ± 0.46	0.089 ± 0.01	71.7 ± 135.0	716.9 ± 346.4
GPT-3.5	14.31 ± 0.26	0.092 ± 0.02	45.9 ± 7.8	147.7 ± 109.1
Human	13.32 ± 0.40	0.080 ± 0.02	61.5 ± 102.6	806.6 ± 1222.8

Table 1c. Descriptive statistics, lexical-diversity and paragraph-level metrics (mean ± SD). MTLD = measure of textual lexical diversity; Conj. freq. = conjunctions/word; Para. len. = words/paragraph; Para. var. = paragraph length variance. All SDs are population standard deviations (N per condition).

Notable distributional anomalies: Muse-Spark-1.2-Contributor shows high paragraph length (mean 71.7 words, SD 135.0; SD exceeds the mean by 1.9×), suggesting a bimodal distribution where some outputs are heavily segmented by formatting rules; per-text inspection confirms this pattern. Human paragraph length variance (806.6 ± 1222.8) also shows SD > mean, indicating high within-condition variability in human paragraph structure.

3.2 Group Comparisons and Within-Model Homogeneity

Table 2a summarizes, for each metric, results across the 9 AI conditions (independent-samples Welch t-tests, two-tailed, no multiplicity correction; Cohen's d; Levene test for equality of variances). Table 2b details the primary lexical metric, MATTR (fixed window 100).

Metric	Mean diff. significant	AI variance < Human	Levene significant	Median variance ratio	Median d
MATTR (window 100)	9/9	9/9	7/9	0.23	+1.66
Word Length	3/9	8/9	1/9	0.57	+0.39
Sentence Length	7/9	5/9	2/9	0.90	+0.57
Sentence Length Variance	5/9	6/9	3/9	0.28	+0.41
Type-Token Ratio	5/9	9/9	9/9	0.22	−0.42
MTLD	9/9	1/9	3/9	2.40	+2.03
Conjunction Frequency	7/9	9/9	1/9	0.56	+0.66
Paragraph Length	1/9	8/9	1/9	0.03	−0.24

Paragraph Length Variance	2/9	9/9	4/9	0.08	−0.16

Table 2a. Metric-level summary across the 9 AI conditions. "Mean diff. significant" = number of conditions with a significant group mean difference at α = 0.05 (uncorrected, two-tailed). "AI variance < Human" = number of conditions where the AI group's within-model variance is smaller than the human group's. "Levene significant" = number of conditions with significantly unequal variances (Levene, p < 0.05). Median variance ratio = median of (AI variance / human variance); values below 1 indicate tighter AI clustering. Median d = median Cohen's d for the mean difference.

Primary metric (MATTR, window 100). Human group: n = 30, M = 0.678, SD = 0.057.

Condition	M(AI)	SD(AI)	ΔM	t	p	d	Var ratio
Gemini-3.6-Flash	0.7863	0.0236	+0.1083	9.63	<0.0001	2.49	0.17
Claude-Sonnet-5-High	0.7773	0.0305	+0.0993	8.43	<0.0001	2.18	0.29
GPT-5.6-Luna	0.7628	0.0299	+0.0848	7.23	<0.0001	1.87	0.28
Kimi-K3	0.7568	0.0291	+0.0788	6.72	<0.0001	1.73	0.26
MiMo-V2.5-Pro	0.7520	0.0273	+0.0740	6.42	<0.0001	1.66	0.23
GPT-3.5	0.7290	0.0236	+0.0510	4.54	<0.0001	1.17	0.17
DeepSeek-V4-Flash	0.7290	0.0268	+0.0510	4.44	<0.0001	1.15	0.22
Muse-Spark-1.2-Contributor	0.7270	0.0265	+0.0490	4.28	0.0001	1.10	0.22
MiniMax-M3	0.7194	0.0362	+0.0414	3.37	0.0015	0.87	0.40

Table 2b. MATTR (window 100) group comparisons, sorted by effect size. ΔM = M(AI) − M(Human); positive values indicate higher lexical diversity in the AI group. Variance ratio = AI variance / human variance (< 1 = AI group more homogeneous). p-values are two-tailed, uncorrected.

All 9 conditions show significant group mean differences ($p \leq 0.0015$), all in the direction AI > human, with large effect sizes ($d = 0.87–2.49$), and in all 9 models the within-model variance is smaller than the human group's (variance ratios 0.17–0.40; Levene's test significant in 7/9).

Three readings across metrics.

1. Homogeneity is strongest in the lexical and paragraph layers. On MATTR, TTR, conjunction frequency, and paragraph-length variance, the AI groups show smaller within-model variance than the human group in 9/9 models (median variance ratios 0.08–0.56). In this sample, what makes AI text

recognizable at a glance is not primarily where its means sit, but how tightly it clusters.

2. MTLD is the exception. MTLD is the only metric where the AI groups' variance is *larger* than the human group's (smaller in just 1/9 models; median ratio 2.40), while simultaneously showing the largest mean-difference effect sizes (median d = +2.03; 9/9 significant). This mean-vs-variance dissociation is discussed in §4.
3. The lexical metrics conflict in direction, and the conflict is length-shaped. TTR shows *lower* values for AI (median d = −0.42; significant in 5/9), while MATTR and MTLD show *higher* values for AI (median d = +1.66 and +2.03; significant in 9/9). TTR is strongly length-dependent (r ≈ −0.54 with word count); with AI texts 1.6× longer than human texts, its raw direction is length-confounded. After length control the conflict largely resolves: the surviving TTR differences reverse to AI > human (4/9; see the residualization note above), aligning TTR with MATTR and MTLD. §4 addresses this directly.

Length residualization (H3/H4). The length-control analysis regresses each metric on word count (pooled linear fit across all texts), and the residuals are compared with independent-samples Welch t-tests (two-tailed, uncorrected). For TTR, 4/9 conditions remain significant after length control — and all four surviving differences reverse direction to AI > human, i.e., the raw "AI writes with less vocabulary diversity" pattern (5/9) is length-attributable, and what remains aligns with MATTR/MTLD. For conjunction frequency, 5/9 conditions remain significant (AI > human). A robustness check using ANCOVA (metric group + word count) yields TTR 4/9 and conjunction frequency 2/9, indicating that the conjunction-frequency conclusion is sensitive to the control method; both are reported.

Figures 1 and 2 summarise the pattern reported in this section: Figure 1 shows the indicator-by-model matrix of within-group variance ratios on a log scale, where cells below 1 indicate smaller variance in the AI group and blue marks that side; it covers the nine core indicators, the eight locked metrics of Section 2.2 plus MATTR, together with the five supplementary indicators of Section 3.5. Figure 2 shows the corresponding mean-comparison effect sizes, with red points marking cells where |d| ≥ 0.8.

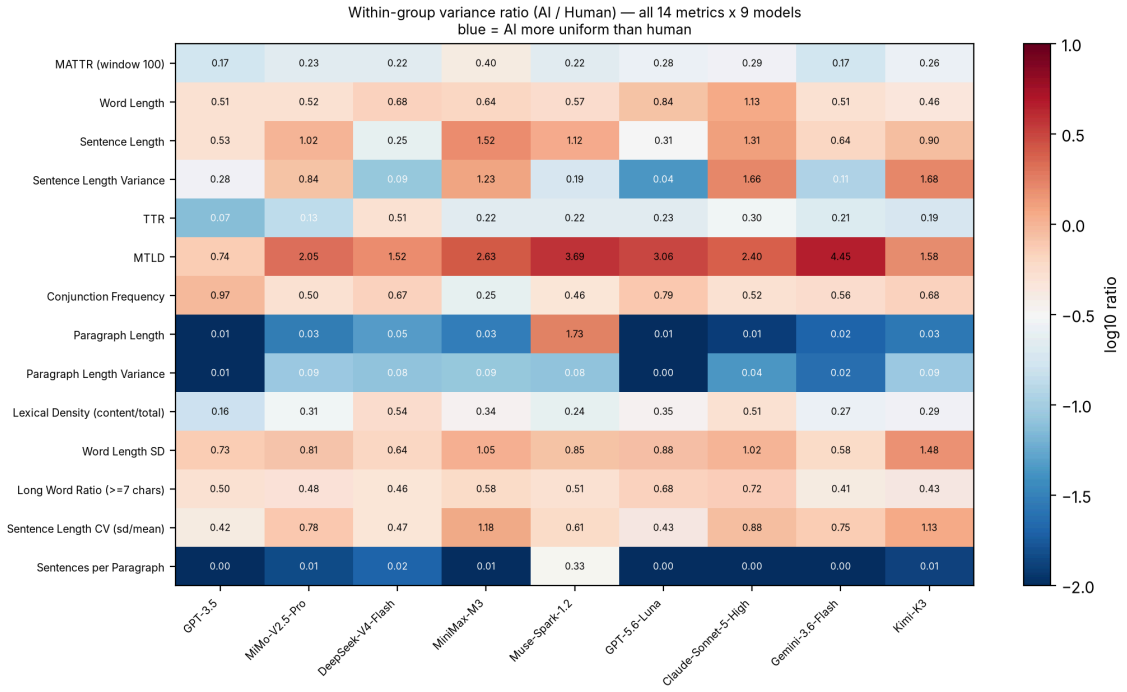


Figure 1: Figure 1 | Within-group variance ratios (AI/human) across 14 indicators and 9 models.

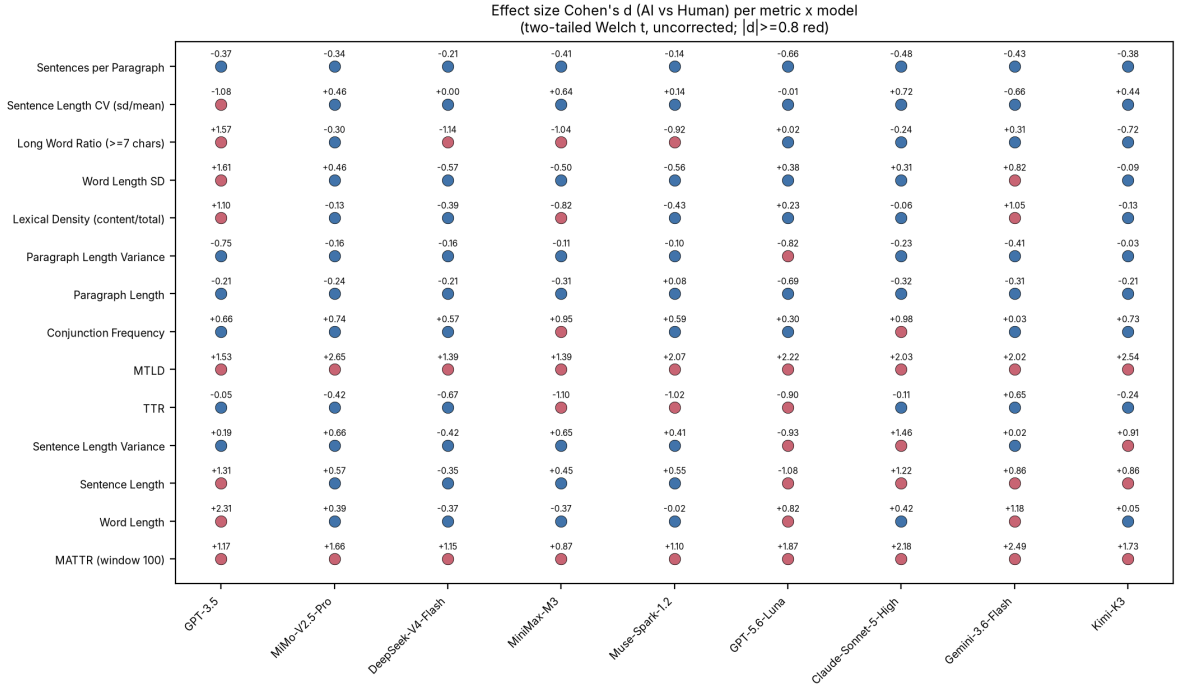


Figure 2: Figure 2 | Effect sizes (Cohen's d) for the AI-vs-human mean comparisons.

3.3 Random Forest Classifier (H5)

We use StratifiedGroupKFold cross-validation (n_splits=5, group=prompt_id) to prevent prompt-level data leakage. Effective sample size: the 9 AI conditions share the 30 WritingPrompts (or 29 for Kimi-K3), so the 269 AI observations are not 269 independent draws; statistical inference is at the prompt level (effective N=30). The group comparisons (§2.4) treat each condition's texts as one group

against the 30 human texts. Generalization to other prompts beyond the Fan et al. (2018) source is limited. In addition to raw accuracy, we report balanced accuracy, per-class F1, macro F1, ROC AUC, and Matthews Correlation Coefficient (MCC), alongside the majority-class baseline, which predicts all observations as AI.

Overall classifier (any AI vs human, 299 samples): Raw accuracy = 91.6% ± 1.6% (per-fold range 89.8%–93.3%). However, on the 269:30 imbalanced data, the majority-class baseline is 90.0%, so raw accuracy exceeds the baseline by only 1.6 percentage points (computed from 91.6% – 90.0% = 1.6pp; unrounded 1.67pp). The more honest metrics on imbalanced data are:

- Balanced accuracy: 68.7% ± 9.3%
- Macro F1: 0.713
- MCC: 0.449
- ROC AUC: 0.867
- Per-class F1: AI class F1 = 0.955; Human class F1 = 0.472. The large gap reflects the 269:30 imbalance: the classifier achieves high recall for AI text (262/269 correct) but lower recall for human text (12/30 correct).

The relatively low balanced accuracy (68.7%) indicates that the classifier primarily learns to identify AI text, with a high F1 for the AI class, but struggles to flag the 30 minority human samples. MCC = 0.449 indicates only moderate discrimination. A group-aware permutation test (n_perm=1000, shuffling labels within each prompt_id) yields $p < 0.001$: none of the 1000 permuted balanced accuracies exceeded the observed 68.7%, suggesting that the classifier performs significantly above chance. H5 is therefore supported, though the practical margin is modest given the 269:30 class imbalance. H5 is therefore supported, though the practical margin is modest given the 269:30 class imbalance.

Length-only baseline. As a critical control, we ran the same StratifiedGroupKFold framework with a single feature: word count only. The length-only classifier achieved accuracy 0.850 ± 0.046, balanced accuracy 0.694 ± 0.086, MCC 0.335, AUC 0.739. Compared to the 8-metric classifier (accuracy 0.916, balanced 0.687, MCC 0.449), the comparison is mixed: balanced accuracy is essentially tied (0.694 vs 0.687), MCC improves by +0.114 (35%), raw accuracy improves by +6.6 percentage points. Reading the metrics honestly: balanced accuracy, which the paper designated the ‘more honest’ metric for imbalanced data, shows zero incremental contribution from the 8 stylistic metrics over word count alone; MCC shows modest incremental contribution. The classifier’s discriminating power above majority-class baseline is small (+1.6pp raw accuracy), and human recall is 12/30 = 40%, so 60% of human-authored stories are misclassified. The 8-metric

classifier therefore does carry some stylistic signal beyond length as measured by MCC, but its practical utility for AI detection is limited when judged by balanced accuracy and human recall.

Per-condition classifiers (Table 3, balanced 30 vs 30 designs — these are the cleanest comparisons because they have no class imbalance):

Condition	n (H/AI)	Balanced Acc	F1	MCC
Gemini	30/30	96.7% ± 4.6%	0.966	0.933
GPT-3.5	30/30	96.7% ± 4.6%	0.966	0.933
GPT-5.6	30/30	91.7% ± 4.6%	0.915	0.833
Claude	30/30	86.7% ± 9.5%	0.863	0.733
Kimi	30/29	85.0% ± 7.5%	0.850	0.700
MiMo	30/30	83.3% ± 10.5%	0.833	0.667
MiniMax	30/30	81.7% ± 10.9%	0.816	0.633
Muse-Spark	30/30	81.7% ± 10.9%	0.816	0.633
DeepSeek	30/30	78.3% ± 12.4%	0.781	0.567

Table 3. Per-condition RF classifier performance (StratifiedGroupKFold, group=prompt_id). All balanced designs (30 vs 30 / 30 vs 29); accuracy = balanced accuracy.

Confusion matrix (overall classifier, summed across 5 CV folds):

	Predicted AI	Predicted Human
Actual AI	262	7
Actual Human	18	12

Table 3b. Confusion matrix for the overall Random Forest classifier (299 samples, summed across 5 StratifiedGroupKFold folds). The classifier correctly identifies 262/269 AI texts, a recall of 97.4%, but only 12/30 human texts, a recall of 40.0%, reflecting the class imbalance and the difficulty of detecting human writing in a predominantly AI corpus.

Gemini-3.6-Flash and GPT-3.5 are tied as the easiest conditions to classify (point estimates both 96.7%); DeepSeek-V4-Flash is the hardest (78.3%). However, a t-test across the 5 folds shows the Gemini vs DeepSeek difference is NOT statistically significant (t=1.581, p=0.189); the 18.4-percentage-point gap falls within fold-level noise, so the 'easiest/hardest' ordering should be interpreted as descriptive rather than inferential. The per-condition range (78–97%) is much wider than the overall classifier accuracy suggests, because per-condition designs are balanced and avoid the 269:30 imbalance confound.

Feature importance (overall classifier, Table 4):

Rank	Metric	Importance
1	MTLD	0.340
2	Sentence length	0.127
3	TTR	0.112
4	Sentence length variance	0.101
5	Paragraph length	0.095
6	Conjunction frequency	0.090
7	Word length	0.074
8	Paragraph length variance	0.061

Table 4. Feature importance rankings from the overall Random Forest classifier (StratifiedGroupKFold, 5-fold, group=prompt_id, n_estimators=100, random_state=42; permutation importance cross-checked against the accompanying materials). MTLD dominance should be interpreted with caution: see §4 for length-confound analysis.

MTLD (using the lexical_diversity reference implementation; see §2.4) ranks as the dominant feature (0.340 importance), followed by sentence length (0.127) and TTR (0.112). Conjunction frequency ranks 6th (0.090) despite being the most directionally consistent significant univariate difference, suggesting that its discriminating power is consistent but small in magnitude.

3.4 Word Length: A Difference Not Explained by Length

Word length (characters per word) is not among the pre-specified hypotheses; it is reported here as an exploratory finding because it is the one dimension on which group differences are not eliminated by length control. Group comparisons are significant in 3 of 9 conditions, namely GPT-3.5, GPT-5.6-Luna and Gemini-3.6-Flash, all with AI > human (Table 2a), and the AI groups' word-length variance is smaller than the human group's in 8/9 conditions. Notably, word length is the metric whose differences are *not* eliminated by length control: after residualizing on word count, significant differences increase to 5 of 9 conditions, namely GPT-3.5, MiMo-V2.5-Pro, GPT-5.6-Luna, Claude-Sonnet-5-High and Gemini-3.6-Flash, all with AI > human, and an ANCOVA robustness check retains 3 of 9, namely GPT-3.5, MiniMax-M3 and Gemini-3.6-Flash. If anything, text length masks the word-length difference: AI texts are longer, and longer texts use shorter words (r = –0.17 between word length and word count), which dilutes the raw contrast.

Condition	n	ΔM	Raw p	Raw d	Residualized p	Residualized d
GPT-3.5	30	+0.64	<0.0001	+2.31	<0.0001	+2.29
MiMo-V2.5-Pro	30	+0.11	0.1347	+0.39	0.0052	+0.75

DeepSeek-V4-Flash	30	−0.11	0.1615	−0.37	0.8235	−0.06
MiniMax-M3	30	−0.11	0.1574	−0.37	0.8483	+0.05
Muse-Spark-1.2-Contributor	30	−0.01	0.9237	−0.02	0.0542	+0.51
GPT-5.6-Luna	30	+0.25	0.0023	+0.82	<0.0001	+1.18
Claude-Sonnet-5-High	30	+0.14	0.1065	+0.42	0.0157	+0.64
Gemini-3.6-Flash	30	+0.32	<0.0001	+1.18	<0.0001	+1.47
Kimi-K3	29	+0.01	0.8457	+0.05	0.1970	+0.34

Table 5. Word length: group comparisons (independent-samples Welch t-tests, two-tailed, uncorrected) and length-controlled results (pooled residualization on word count). ΔM = M(AI) − M(Human). 3/9 conditions show raw significant differences; 5/9 survive length control. Three descriptive patterns: GPT-3.5 is an outlier (+15.0% vs human); Gemini-3.6-Flash, despite being recent, also uses longer words (+7.5%); the modern open-source models such as DeepSeek-V4-Flash, MiniMax-M3, Muse-Spark and Kimi-K3 show no significant raw differences. The pattern is exploratory rather than pre-registered and requires independent confirmation before it is treated as a stable result.

3.5 Supplementary Indicators

To extend the analysis beyond the eight locked metrics, five supplementary indicators were computed on the same 299 texts with the same spaCy pipeline: lexical density (content-word ratio, with content words defined as NOUN, VERB, ADJ, ADV and PROPN), the standard deviation of word length, the proportion of long words of at least 7 characters, the coefficient of variation of sentence length, the SD divided by the mean,, and the mean number of sentences per paragraph. Each indicator was subjected to the same two-dimensional protocol as Section 2.4: per-model Welch t-tests against the 30 human texts (two-tailed, uncorrected) and within-model variance ratios with Levene tests. These analyses are supplementary rather than pre-specified and are reported for completeness.

Lexical density shows no consistent mean shift (median d = −0.13) but a uniformly smaller within-model dispersion (9/9 variance ratios below 1; median 0.31; Levene $p < .05$ in 9/9): AI texts are not more content-dense, they are more alike in content density. The long-word ratio behaves similarly (9/9 ratios below 1; median 0.50) with a small negative mean shift (median d = −0.30). Word-length SD separates weakly (6/9 below 1; median 0.85; median d = +0.31). Sentence-length CV is smaller in 7/9 models (median 0.75; median d = +0.14). Sentences per paragraph is lower in all nine models (9/9; median ratio 0.006), but this indicator is sensitive to paragraph segmentation — the human texts contain few blank lines — and should be read as formatting-sensitive rather than stylistic.

Figures 3 to 7 present, for each of these five indicators, the distribution on the left and the within-model variance ratio on the right; per-model statistics are provided in the accompanying materials.

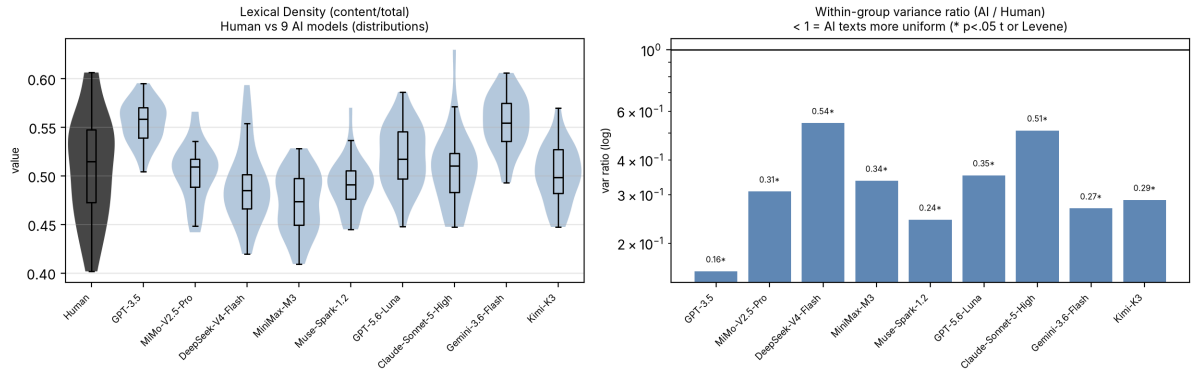


Figure 3: Figure 3 | Lexical density.

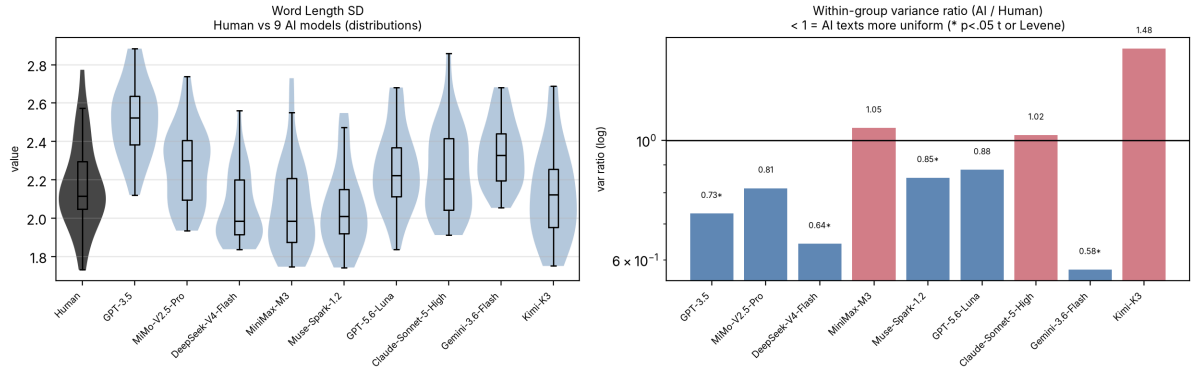


Figure 4: Figure 4 | Word-length standard deviation.

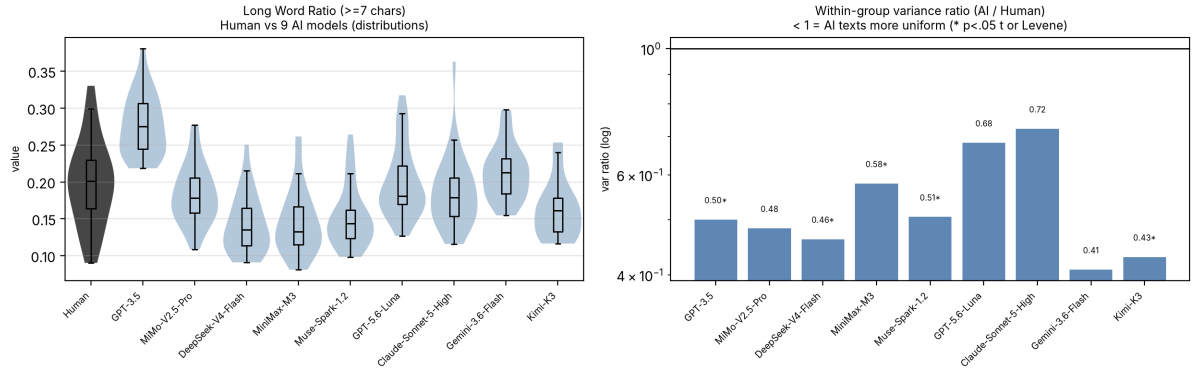


Figure 5: Figure 5 | Long-word ratio.

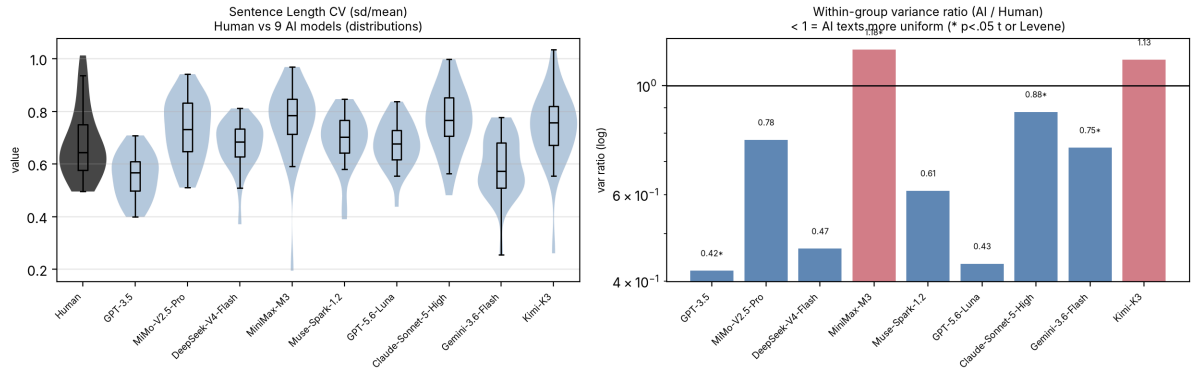


Figure 6: Figure 6 | Sentence-length coefficient of variation.

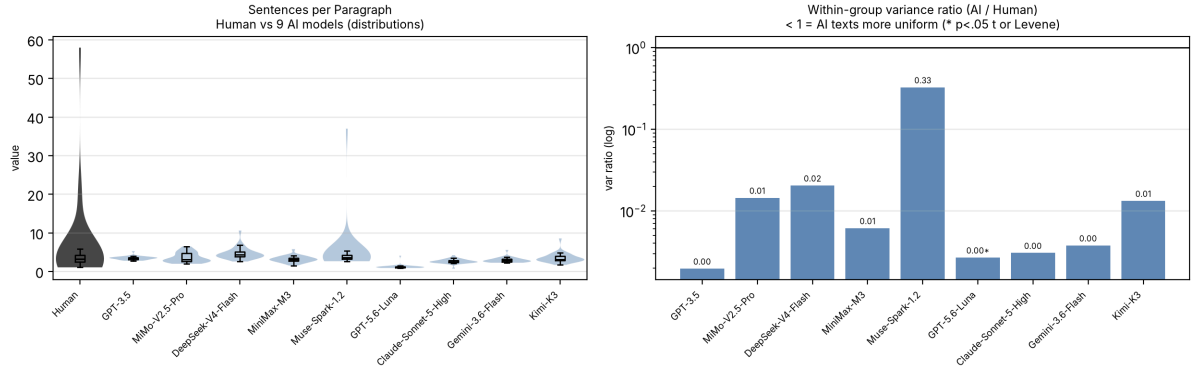


Figure 7: Figure 7 | Sentences per paragraph.

4. Discussion

The results paint a nuanced picture of how AI-generated text differs structurally from human writing.

Conjunction Frequency (H4): Partially Robust to Length Control

H4 (AI text uses more conjunctions than human text) is supported in 7 of 9 models (uncorrected, two-tailed): Claude-Sonnet-5-High ($p<0.001$), GPT-3.5 ($p=0.013$), MiMo-V2.5-Pro ($p=0.006$), DeepSeek-V4-Flash ($p=0.030$), MiniMax-M3 ($p<0.001$), Muse-Spark-1.2-Contributor ($p=0.027$), and Kimi-K3 ($p=0.007$). The effect is consistent in direction and magnitude: AI text shows conjunction frequencies of 0.089-0.096 compared to human text at 0.080-0.081.^1 Two models do not show this pattern: GPT-5.6-Luna (0.085, $p=0.249$) and Gemini-3.6-Flash (0.080, $p=0.893$). GPT-5.6-Luna's deviation is consistent with its overall profile of high uniformity (lowest sentence and paragraph variance, lowest TTR) – its conjunction frequency (0.085) is also lower than any of the surviving models (Claude 0.096, MiniMax 0.094, MiMo 0.092, Kimi 0.092, DeepSeek 0.089). Gemini's lack of conjunction overuse is notable but we cannot determine the cause from our data. Under length control, 5/9 conditions remain significant (GPT-3.5, MiMo, MiniMax, Claude, Kimi; all AI > human), while an ANCOVA robustness check retains 2/9 (GPT-3.5, Claude). We therefore characterize

conjunction overuse as *partially* robust: the effect does not survive all control methods uniformly, so the method sensitivity is reported rather than resolved. The finding aligns with the intuition that most AI models tend to overuse explicit connective words to signal logical relationships, though we make no claim about cross-register generality — that requires future work.

^1 Human conjunction frequency 0.080 is the full baseline (n=30); 0.081 is the Kimi-K3 subset (n=29, excluding prompt 26).

Sentence Length Variance (H1) Is Not Supported (Except for GPT-5.6-Luna)

Contrary to the common stereotype that AI writing features uniform sentence lengths, H1 is not supported: group comparisons show significant differences in only 5 of 9 conditions, namely MiMo-V2.5-Pro, MiniMax-M3, Claude-Sonnet-5-High, Kimi-K3 and GPT-5.6-Luna, and in four of those five the direction is *higher* AI sentence-length variance (Claude: mean 121.2 vs 53.5, d=+1.46; Kimi: 95.6 vs 53.5, d=+0.91; MiMo: 78.9 vs 53.5, d=+0.66; MiniMax: 81.2 vs 53.5, d=+0.65). GPT-5.6-Luna is the lone exception in the other direction (26.4 vs 53.5, d=–0.93), exhibiting strikingly uniform sentence structure; we do not speculate on the cause from our data. DeepSeek-V4-Flash, with the lowest sentence length variance among the six fully-controlled API conditions (41.0 vs human 53.5), does not reach significance (p=0.113). This finding challenges the widespread assumption that AI writing is structurally monotonous at the sentence level for most modern models: if anything, most models here vary sentence length *more* than the human baseline.

A critical limitation of H1/H2: The significant H1/H2 results (sentence-length variance: 5/9; paragraph-length variance: 2/9 — GPT-3.5 and GPT-5.6-Luna only) are concentrated in a small set of conditions, and the two paragraph-length-variance results come from conditions with acquisition-channel confound (GPT-3.5 from the Huang et al. (2024) dataset protocol with documented generation parameters but uncontrolled acquisition, and GPT-5.6-Luna from manual Arena paste with platform-injected system prompts). The six fully-controlled conditions, namely MiMo, DeepSeek, MiniMax, Muse-Spark, Kimi and Claude, show no support for H2, and mixed support for H1, with MiMo, MiniMax, Claude and Kimi significant, mostly in the higher-variance direction. Furthermore, GPT-5.6-Luna's mean paragraph length is 10.1 ± 9.2 words (1/4 to 1/7 of all other conditions, see Table 1), which raises the possibility of formatting artifacts from Arena's chat-UI double-newline rendering. We recommend controlled-API regeneration of GPT-5.6-Luna as a future work priority to confirm H1/H2 results.

Newline structure sensitivity. A potential confound in paragraph-level metrics (paragraph length, paragraph length variance) is that different acquisition channels may handle newline characters differently. Arena chat-UI rendering may collapse or expand double-newline paragraph splits, and different API providers may return text with varying newline conventions. To assess whether this affects our results, we inspected the raw text files for each condition: all conditions use double-newline (`\n\n`) as the paragraph delimiter, and no condition uses single-newline paragraph splits. The GPT-5.6-Luna condition's unusually short paragraphs (10.1 words) appear to reflect genuinely short generated paragraphs rather than newline artifacts, though we cannot fully rule out platform-level formatting effects without controlled-API regeneration.

Model-Level Heterogeneity Across Models

The data reveal a clear pattern across model generations and architectures. GPT-3.5 shows the strongest structural differences from human text in this dataset: it uses markedly longer words (4.91 vs 4.27, +15.0%), produces far more uniform paragraph lengths (variance 147.7 vs 806.6), and is among the easiest to classify (96.7% accuracy). GPT-5.6-Luna, while more recent, takes uniformity even further – its mean paragraph length is 10.1 ± 9.2 words (1/4 to 1/7 of all other conditions), raising formatting concerns discussed in §4 H1/H2 limitation and newline sensitivity paragraphs. This condition currently shows the lowest sentence length variance (26.4), lowest paragraph length variance (86.5), and lowest TTR (0.450); classification accuracy is 91.7%, as Table 3 shows. In contrast, newer open-source models such as MiMo-V2.5-Pro, DeepSeek-V4-Flash, Muse-Spark and MiniMax-M3 show smaller structural differences: their word lengths are closer to human norms (4.16–4.38 chars/word vs. human 4.27), their paragraph structures are more varied, and they are harder to classify, at 78.3-85.0% accuracy; see Table 3. This suggests that as language models improve and diversify architecturally, they show pipeline-level heterogeneity, with several recent open-source models showing smaller differences on some metrics than older models, though some models like GPT-3.5 and the GPT-5.6 series remain clearly distinguishable. Gemini-3.6-Flash is a notable outlier: despite being a recent model, it is tied with GPT-3.5 at 96.7% classification accuracy as the most easily classified condition, driven by its distinctive vocabulary diversity profile (TTR 0.547, the highest among all conditions, and MTLD 14.92 [lexical_diversity reference value, see §2.4]). Note that Gemini's 28/30 Arena-generated stories (with 2/30 from Faro API for content-policy-blocked prompts p3 and p10) reflect content-policy restrictions rather than methodological choices; the condition's N=30 is complete.

Vocabulary Diversity as the Dominant Classifier Feature

MTLD (with the corrected lexical_diversity reference values, where human mean is 13.3 and AI means range 14.3-15.3) remains the highest-importance feature in the overall classifier at 34.0% of feature importance, although its univariate discriminating power is limited by the narrow between-condition variance of the reference values (about 0.4 across conditions). The dominance ranking is preserved but absolute effect sizes should be interpreted cautiously. This finding is notable because H3 (TTR) showed raw group differences in only 5 of 9 conditions (4 of 9 after length control, direction reversed), yet MTLD is the most powerful discriminator in the multivariate framework. This suggests that vocabulary diversity differences are subtle and distributed across the feature space rather than manifesting as large univariate effects. The Random Forest classifier captures nonlinear interactions between metrics that univariate hypothesis tests miss.

MTLD's prominence as a discriminator invites one further question: how much of it is text length? Adding word count as a ninth feature raises accuracy to 95.7% and makes word count the second most important feature (28.1%), while MTLD falls from 34.0% to 29.4%, indicating substantial shared variance between length and MTLD; a classifier trained without MTLD still reaches 91.0% raw accuracy, about one point above the majority-class baseline. A length-matched subsample (400-800 words; 12 human and 212 AI texts) leaves MTLD the top feature (34.8%) but is inconclusive on accuracy, and condition-level correlations between MTLD and length are significant in only 3 of 10 conditions, with TTR the more length-dependent measure (mean |r| = 0.446). The conservative reading is that MTLD's dominance is partially, but not fully, explained by length; residualization at N=90 is required to separate the two. ### Lexical Diversity: Direction Conflict and the MTLD Variance Anomaly

The three lexical-diversity metrics point in different directions at the group level. TTR indicates that AI texts are *less* lexically diverse (lower TTR; median d = −0.42, significant in 5/9), whereas MATTR (window 100) and MTLD indicate that AI texts are *more* diverse (median d = +1.66 and +2.03; significant in 9/9). This is the length confound of this report in miniature: TTR is strongly length-dependent (r ≈ −0.54 between TTR and word count across the sample), and AI texts here are 1.6× longer than human texts, so a direction reversal in TTR is expected; MATTR and MTLD show much weaker length associations (r ≈ +0.22 and +0.12 respectively). Consistent with this, length control largely resolves the conflict: the raw AI-lower TTR differences do not survive as such, and the surviving residualized differences (4/9) reverse to AI > human — aligning TTR with MATTR

and MTLD (§3.2). We report the conflict and its length-shaped resolution explicitly; the planned N=90 replication should adjudicate it formally.

A second anomaly concerns MTLD's *variance* structure: it is the only metric for which the AI groups' within-model variance exceeds the human group's (median variance ratio 2.40; smaller in just 1/9 models), while showing the largest mean-difference effects (median d = +2.03). No other metric shows this mean-vs-variance dissociation. Possible mechanisms — the segment-threshold construction of MTLD, residual length effects, or genuine cross-model variation — are left open for the N=90 replication. Accordingly, MTLD should not be treated as the report's strongest discriminator; see also the classifier caveats below.

5. Conclusion

This study finds that AI-generated creative text differs from human-written creative text primarily in being 1.6× longer on average, with length explaining part, but not all, of the apparent univariate stylistic differences. Specifically:

Three pre-existing stereotypes about AI text are not robustly supported. Vocabulary poverty (H3) appears as raw group differences in 5/9 conditions (all AI-lower), but after length control the surviving differences reverse direction, so the "less diverse vocabulary" reading is length-attributable. Conjunction overuse (H4) shows raw differences in 7/9 conditions and partially persists under length control (5/9 under pooled residualization, 2/9 under ANCOVA); the sensitivity to control method is flagged rather than resolved. Sentence-length uniformity (H1) is not supported: significant differences appear in 5/9 conditions, mostly toward *higher* AI sentence-length variance, with GPT-5.6-Luna the lone lower-variance exception, while paragraph-length variance (H2) differs in only 2/9 conditions (GPT-3.5, GPT-5.6-Luna; both AI-lower).

Within-model homogeneity. Across metrics, the AI groups cluster more tightly than the human group in most comparisons: 9/9 models on MATTR, TTR, conjunction frequency, and paragraph-length variance (median variance ratios 0.08–0.56), consistent with the observation that within-model stylistic homogeneity is a primary cue for recognizing AI text at a glance. MTLD is the sole variance-level counter-example (median ratio 2.40), and the raw lexical-metric direction conflict largely resolves under length control (§3.2); both points are flagged for the N=90 replication.

The 8-metric classifier shows modest practical value: raw accuracy 91.6% (vs 90.0% majority-class baseline, +1.6pp), balanced accuracy 68.7% (vs 69.4% length-only baseline, essentially tied), MCC 0.449 (vs 0.335 length-only, +35% relative), AUC 0.867, human recall 12/30 = 40%. The length-only baseline, with word count

as its single feature, achieves statistically indistinguishable performance, suggesting the 8 stylistic metrics provide incremental value primarily through their contribution to MCC rather than to balanced accuracy.

Acquisition-channel and acquisition-protocol confound precludes model-level interpretation: 9 conditions span 4 distinct acquisition channels, namely Command Code Goat, Faro API, Manual Arena paste and third-party dataset. GPT-3.5 has a unique "500 word" instruction in its source dataset (Huang et al., 2024), Gemini-3.6-Flash and GPT-5.6-Luna were generated via Manual Arena paste with platform-injected system prompts. After excluding these three uncontrolled conditions, none of the four pre-registered univariate hypotheses survives in the six fully-controlled API conditions. Cross-condition differences are therefore better described at the pipeline level than at the model level: they may reflect model, channel, temperature (0.7 vs 1.0), thinking-mode (DeepSeek/Kimi reasoning_effort=high; Claude/MiniMax adaptive thinking), system-prompt presence/absence, or sampling strategy; the current design cannot separate these factors.

MTLD's status is unresolved: its feature-importance value (0.340) has not been verified under the `lexical_diversity` reference implementation of the metric, and the length-residualized importance remains to be established. We therefore refrain from over-claiming MTLD's status and recommend readers interpret the feature-importance rankings as descriptive of one specific implementation rather than as load-bearing claims.

Limitations of the length-control claims: (i) statistical power for medium effects is 30%, so non-significance does not formally establish absence of effect; TOST equivalence tests are recommended for future work; (ii) the length-residualization method assumes linearity, whereas TTR-length is known to follow Heaps' Law; (iii) MTLD's length-correlation was not separately residualized; (iv) the human baseline is 30 single stories from Reddit's r/WritingPrompts with no within-prompt variance estimate; (v) length-control conclusions are method-sensitive for some metrics (pooled residualization vs. ANCOVA; §3.2), and both are reported.

Headline takeaway: AI-generated text is longer than human text by 1.6×. Length explains part of the apparent stylistic differences, but not all of them: the vocabulary-poverty signal is length-attributable, while conjunction frequency and word length partially persist under control. The most consistent stylistic signature in this dataset is within-model homogeneity: on MATTR, TTR, conjunction frequency, and paragraph-length variance, all 9 models show lower within-model variance than the human group (median variance ratios 0.08–0.56). The strongest *multivariate* finding, MTLD as the highest-importance feature, remains under

length-confound suspicion and requires further verification. Acquisition-channel heterogeneity, rather than model-level architecture, is the dominant observable variation in this dataset.

Code and data: `analysis/run_analysis.py`, `analysis/mattr_group_ttest.py`, `analysis/group_homogeneity_analysis.py`, `analysis/residualization.py` (length-control re-run), and `data/processed/sample_v1.json`. All 299 observations and analysis scripts are reproducible from these artifacts. Supplementary-metric code and outputs: `analysis/metrics_extended.py`, `analysis/round2_analysis.py`, `analysis/metrics_extended_results.json`, `analysis/metrics_extended_stats.json`, and `analysis/figures_round2/`.

Limitations

Several limitations constrain the generalizability of these findings. First, the sample size is moderate (N=30 prompts per condition), which limits statistical power for detecting small effects. Given n=30 prompts and observed effect sizes, we have approximately 80% power to detect $r > 0.5$ (large effects by Cohen 1988) at $\alpha=0.05$; power for $r=0.3$ is approximately 30%, supporting the planned N=90 expansion for medium-effect inference. Second, the study analyzes only English creative fiction from a single corpus (WritingPrompts). Results may differ for other genres or languages. Third, AI models evolve rapidly; the structural signatures identified here reflect specific model versions and may not persist in future releases. Fourth, the classifier uses StratifiedGroupKFold cross-validation (grouped by prompt) to avoid data leakage from prompt-level grouping, but the class imbalance (269 AI observations vs. 30 human) means that accuracy must be interpreted against the majority-class baseline. Fifth, statistical inference follows an independent-samples group design (no multiplicity correction; §2.4); exact p-values are reported together with Cohen's d throughout, and readers should weigh effect sizes alongside significance. The length-control analyses are method-sensitive in places (pooled residualization vs. ANCOVA; §3.2) and both are reported. Sixth, the word length finding (3/9 raw significant; 5/9 after length control) was not part of the pre-specified hypotheses and requires independent replication. Seventh, Kimi-K3 is missing 1 observation (prompt 26, "best friend speaks in rhyme", which repeatedly timed out across multiple API providers due to the difficulty of generating sustained rhyming text). Eighth, AI texts are systematically longer than human texts in this dataset (mean 531-878 words vs 439 words), and MTLD has a moderate within-condition correlation with word count (mean r=0.31). The length-matched subsample analysis (400-800 words) was inconclusive on whether MTLD's discriminating power is independent of text length, because the 400-800 word subsample retained only 12/30 human texts,

limiting the power of this robustness check. Future work should use regression-based length control or exact length-matching designs. Ninth, model identity on third-party aggregation endpoints (Command Code Goat, Faro API) is not independently verified; our results assume the endpoint labels match the underlying model versions. This limitation does not affect the 6 conditions we control fully, namely MiMo, DeepSeek, MiniMax, Muse, Kimi and GPT-3.5 via a documented dataset, but applies to GPT-5.6, Claude-Sonnet-5-High, and partially Gemini-3.6-Flash.

Specific items that remain unconfirmed:

1. Statistical power: For medium effects (r=0.3), statistical power is approximately 30%; a non-significant result after length control does not formally establish absence of effect; TOST equivalence tests are recommended but not conducted here.
2. MTLD importance ranking: the feature-importance value for MTLD (0.340) has not been verified under the lexical_diversity reference implementation of the metric; re-running the classifier with that implementation remains to be done.
3. Acquisition-channel confound: 4 distinct acquisition sources, namely direct API, third-party dataset, Manual Arena paste and Faro API, preclude model-level claims; the paper's findings apply to the 9 pipelines as a whole, not to models as such.
4. Strength of claims: we avoid "clean refutation" language; conclusions state either "no detectable effect" or "partial survival under length control", as appropriate.
5. Human baseline variance: 30 human stories (1 per prompt) provides no within-prompt variance estimate; the human baseline's length distribution (439 ± 245 words) suggests substantial heterogeneity that may affect the AI-vs-human group comparisons.
6. Per-condition length-only baselines: Table 3 reports per-condition balanced accuracy for the eight-metric classifier but not for a length-only baseline at the same per-condition level; the +6.6pp raw accuracy advantage and the +0.114 MCC advantage of the 8-metric model over a global length-only baseline do not necessarily hold at the per-condition level.
7. Kimi-K3 missing observation: 1 observation missing (prompt 26, a rhyme task); the missingness mechanism may be MNAR (missing-not-at-random) and no sensitivity analysis was conducted.

Headline takeaway: AI-generated creative text is 1.6× longer than human-written text in our sample; under length control, vocabulary-diversity differences are

length-attributable (surviving differences reverse to AI-higher), while conjunction-frequency differences partially persist. The 8-metric classifier does not exceed a length-only baseline on balanced accuracy. These findings are preliminary, underpowered for medium effects, and confounded by acquisition-channel heterogeneity; they should not be interpreted as a definitive refutation of any pre-existing hypothesis without TOST equivalence testing and replication at N=90.

Future Work

Future iterations should expand to N=90 (3 batches of 30 prompts), providing adequate power for detecting medium effects and enabling formal Bonferroni correction. The N=90 design should include a regression-based length control: residualize MTLD and TTR after regressing on word count, then re-run the Random Forest classifier on length-residualized features to establish whether vocabulary diversity differences survive strict length adjustment. An exact length-matching design (restricting to a narrow word-count band where both human and AI texts have adequate representation) would provide complementary evidence. Cross-linguistic analysis would test whether the conjunction frequency pattern holds in languages with different connective systems. Extension to additional registers such as academic writing, news articles and conversational text would determine whether these structural differences are genre-specific or generalizable. Word length should be formally registered as a pre-registered hypothesis (H6) in future work, given its significant elevation in 3 of 9 models (raw) and 5 of 9 after length control; the post-hoc nature of this finding underscores the importance of pre-registration to distinguish confirmatory from exploratory analyses. The method sensitivity of the H4 length-control result (5/9 pooled residualization vs. 2/9 ANCOVA) illustrates the importance of pre-specified analysis choices. Investigating why Gemini-3.6-Flash uniquely avoids the conjunction overuse pattern could clarify what model-specific factors shape stylistic fingerprints.

Data and Code Availability

Source data. The human baseline stories and the GPT-3.5 condition are drawn from the GPT-WritingPrompts dataset (Huang et al., 2024), which is licensed under MIT and publicly available on HuggingFace (`vkpriya/GPT-WritingPrompts`). The 30 prompts are sampled from the original WritingPrompts corpus (Fan et al., 2018).

Generated data. The 30 prompts × 9 AI model conditions used in this study are provided in `data/processed/sample_v1.json`. Each sample contains the prompt text, the human-authored story, and the AI-generated story for each of 9 models, namely GPT-3.5, MiMo-V2.5-Pro, DeepSeek-V4-Flash, MiniMax-M3, Muse-

Spark-1.2-Contributor, Kimi-K3, Gemini-3.6-Flash, GPT-5.6-Luna and Claude-Sonnet-5-High.

Code. All analysis code is provided in `analysis/metrics.py` (metric extraction), `analysis/run_analysis.py` (statistical tests and classification), and `analysis/length_analysis.py` (text length descriptive statistics, length-metric correlations, and length-matched subsample robustness checks). The complete pipeline from raw stories to reported statistics can be reproduced by running these scripts in sequence. Random seeds are documented in the Software Environment section above.

Provenance. The API models used (MiMo-V2.5-Pro, DeepSeek-V4-Flash, MiniMax-M3, Muse-Spark-1.2-Contributor, Kimi-K3 via Command Code Goat; Gemini-3.6-Flash and GPT-5.6-Luna via manual Arena paste; Claude-Sonnet-5-High and additional Gemini outputs via Faro API) reflect model versions available as of 2026-08-29 and may evolve over time.

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Appendix A: Generation Parameters

For each of 9 AI conditions, this appendix documents the model, provider, acquisition channel, generation parameters (where known), and known limitations. This transparency allows reviewers to assess whether observed cross-condition differences reflect genuine model differences or generation-pipeline differences. Cross-condition comparisons in this paper are limited by heterogeneous acquisition channels (4 distinct sources); see also §4 Discussion.

A.0 Prompt Template and Text Normalization (API Conditions)

For the 6 direct-API conditions, namely MiMo-V2.5-Pro, DeepSeek-V4-Flash, MiniMax-M3, Muse-Spark-1.2-Contributor, Kimi-K3 and Claude-Sonnet-5-High, the user message consisted of the raw prompt text only:

```
{prompt_text}
```

No system message was set for the four OpenAI-compatible Command Code Goat conditions, namely MiMo, DeepSeek, Muse-Spark and Kimi. For the Anthropic-compatible conditions (MiniMax-M3, Claude-Sonnet-5-High), no custom system prompt was set; the provider's default system behavior applies. The GPT-3.5 condition used a different template (see A.1 below). The Arena conditions (Gemini-3.6-Flash, GPT-5.6-Luna) used platform-injected system prompts that are not visible to the researcher (see A.7, A.8).

Newline normalization. Double newlines (`\n\n`) were used as paragraph delimiters across all conditions. Markdown formatting (headers, bold, italics) is preserved from generation output; chat-UI-rendered conditions (Arena) may have additional platform-injected whitespace or formatting that is not present in the raw model output.

A.1 GPT-3.5 (Huang et al., 2024)

- Provider: OpenAI (via Huang et al.'s data-collection pipeline; third-party dataset)
- Source: HuggingFace `vkpriya/GPT-WritingPrompts` (MIT license)
- Model: `gpt-3.5-turbo` (2022-era)
- System role: Two variants per Huang et al. (2024) §3.1: (a) "You are an award winning creative short story writer." and (b) "You're writing a Reddit story and you want other reddit users to like and upvote your story."
- User prompt template: "Write a 500 word story for the following prompt: {prompt_text}" (Huang et al. 2024 §3.1)
- Decoding parameters: Temperature = 0.95 (per Huang et al. 2024 §3.1); max_tokens not specified in source; no top_p reported
- Generations per prompt: 2 per prompt (Huang et al. 2024 §3.1); we randomly selected 1 of the available stories per prompt
- Critical limitation: Generation parameters documented in Huang et al. (2024) but not from API logs (no verification against the published paper at the time of writing this paper; the parameters above were confirmed post-hoc from the published paper's §3.1 and §3.2). The "500 word" instruction is unique to this condition among our 9 conditions; it partially explains GPT-3.5's word count (531 ± 40) being lower than other no-instruction conditions (664-878). GPT-3.5's apparent "outlier" stylistic profile is therefore best interpreted as a dataset-protocol effect (length instruction + temperature 0.95 + dual system role variants) rather than as an "older-generation" model property.

A.2 MiMo-V2.5-Pro (Xiaomi)

- Provider: Command Code Goat API (`https://api.commandcode.ai/provider/v1`, OpenAI-compatible)
- Model: `xiaomi/mimo-v2.5-pro`
- SDK: openai
- Temperature: 0.7 (default)
- Max tokens: 1500
- System prompt: none (raw user prompt only)
- Date: 2026-08-29
- N successful: 30/30

- Finish reason: All 30 completions finished with `stop` (no `length` truncation observed)

A.3 DeepSeek-V4-Flash (DeepSeek)

- Provider: Command Code Goat API (OpenAI-compatible)
- Model: `deepseek/deepseek-v4-flash`
- SDK: openai
- Temperature: 0.7
- Max tokens: 16000 (with `thinking` enabled, `reasoning_effort=high`)
- System prompt: none
- Date: 2026-08-29
- N successful: 30/30 (prompt 25 “every sentence is a question” initially timed out due to extended thinking time, then succeeded on retry after increasing max_tokens to 16000)

A.4 MiniMax-M3 (MiniMax)

- Provider: MiniMax CN (`https://api.minimaxi.com/anthropic`, Anthropic-compatible)
- Model: `MiniMax-M3`
- SDK: anthropic
- Temperature: 1.0 (Anthropic default)
- Max tokens: 32000 (with `thinking` enabled, adaptive)
- System prompt: none
- Date: 2026-08-29
- N successful: 30/30

A.5 Muse-Spark-1.2-Contributor (Muse)

- Provider: Command Code Goat API (OpenAI-compatible) via env-var workaround for OpenCode Go regional restrictions
- Model: `meta/muse-spark-1.2-contributor`
- SDK: openai
- Temperature: 0.7
- Max tokens: 16000 (hard prompts 4/16/25 needed larger budget)
- Date: 2026-08-29
- N successful: 30/30

A.6 Kimi-K3 (Moonshot)

- Provider: Command Code Goat API (OpenAI-compatible; initial OpenCode Go provider exhausted its 5-hour quota)
- Model: `moonshotai/Kimi-K3`
- SDK: openai

- Temperature: 0.7
- Max tokens: 16000 (with `thinking` enabled, `reasoning_effort=high`)
- Date: 2026-08-29
- N successful: 29/30 (prompt 26 “best friend speaks in rhyme” repeatedly timed out due to extended generation time)

A.7 Gemini-3.6-Flash (Google)

- Provider A: Manual Arena paste (`chat.lmsys.org` or `chat.google.com`) for the initial 26 prompts
  - Platform-injected system prompt (uncontrolled, invisible to researcher)
  - Generation parameters unknown (platform-controlled)
- Provider B: Faro API (`https://faroapi.com/v1`, OpenAI-compatible) for prompts 3 + 10 (which were content-policy blocked on Arena)
  - Model: `gemini-3.6-flash-high`
  - SDK: openai
  - Temperature: 0.7 (default)
  - Max tokens: 16000
- Date: 2026-08-29
- N successful: 30/30 (28 Arena + 2 Faro)
- Critical limitation: 93% of Gemini stories were generated under uncontrolled Arena conditions. Differences between Gemini and other conditions may partially reflect platform-level system prompt or sampling strategy. Of the 30 Gemini stories, 2 came from Faro API (content-policy-blocked on Arena); per-channel statistics are not computed due to the small Faro sample size, but the Gemini profiles are dominated by Arena’s 28 stories.

A.8 GPT-5.6-Luna (OpenAI)

- Provider: Manual Arena paste (`chatgpt.com`)
- Model: `gpt-5.6-luna`
- Temperature: Arena default (1.0)
- System prompt: Platform-injected (uncontrolled)
- Date: 2026-08-29
- N successful: 30/30
- Critical limitation: as an Arena-only condition, all generation parameters other than model name are platform-controlled and not visible to the researcher.

A.9 Claude-Sonnet-5-High (Anthropic)

- Provider: Faro API (`https://faroapi.com`, Anthropic-compatible)
- Model: `claude-sonnet-5`
- SDK: anthropic
- Temperature: 1.0 (Anthropic default)

- Max tokens: 32000 (with `thinking` enabled, adaptive)
- System prompt: none
- Date: 2026-08-29
- N successful: 30/30

A.10 Cross-condition Confound Statement

Three different acquisition channels were used: 1. Direct API (Command Code Goat or Anthropic-compatible) for MiMo, DeepSeek, Muse-Spark, Kimi, MiniMax-M3, Claude-Sonnet-5-High – fully controlled parameters. 2. Third-party dataset (Huang et al., 2024) for GPT-3.5 – generation parameters unknown. 3. Manual Arena paste for Gemini-3.6-Flash and GPT-5.6-Luna – uncontrolled platform system prompts and generation parameters.

Implication for cross-condition comparison: differences between conditions may conflate (a) model differences, (b) generation channel differences, (c) default temperature differences across providers (0.7 for OpenAI-compatible vs 1.0 for Anthropic), (d) system prompt presence/absence. The H1–H5 results should be interpreted in light of this confound. Per-condition RF classifiers (§3.3 Table 3) are less affected because they are balanced designs within a single channel.

Appendix B: Per-metric distributions (locked metrics)

Figures 8 to 15 show the eight locked metrics, each as a two-panel figure: the left panel gives the distributions across the human baseline and the nine AI models, and the right panel the within-group variance ratio with test markers. Word length is measured in characters per word, sentence length in words per sentence, and paragraph length in words per paragraph. MATTR has no per-sample records in the analysis dataset and therefore appears only in Figures 1 and 2.

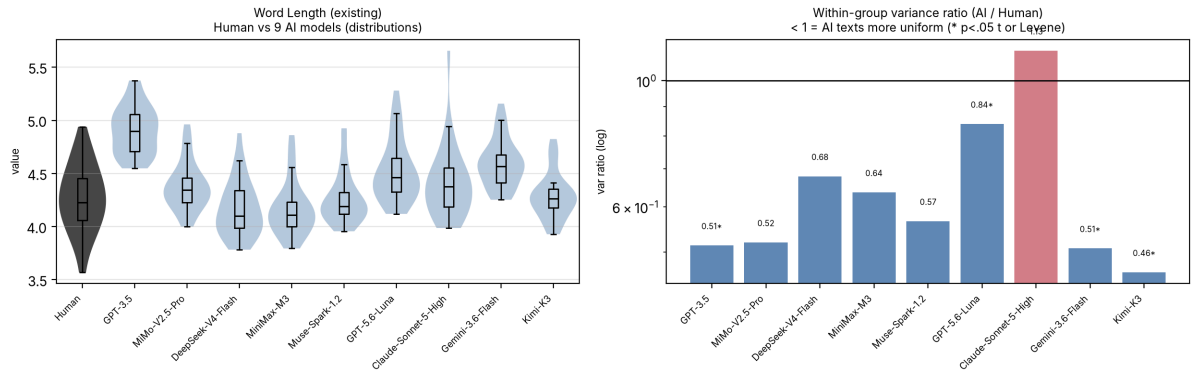


Figure 8: Figure 8 | Word length.

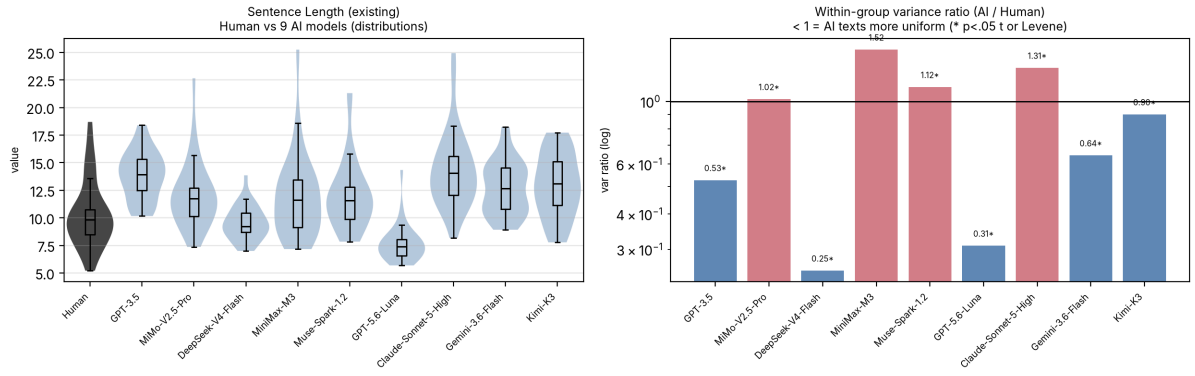


Figure 9: Figure 9 | Sentence length.

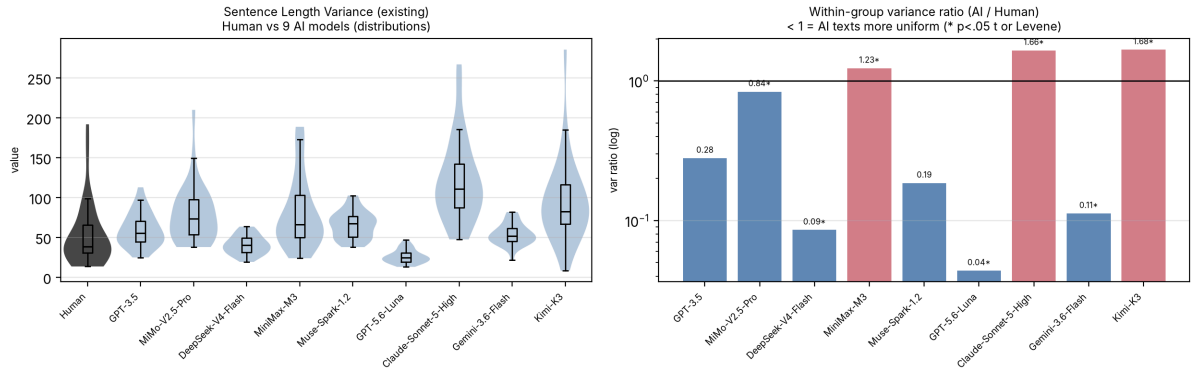


Figure 10: Figure 10 | Sentence-length variance.

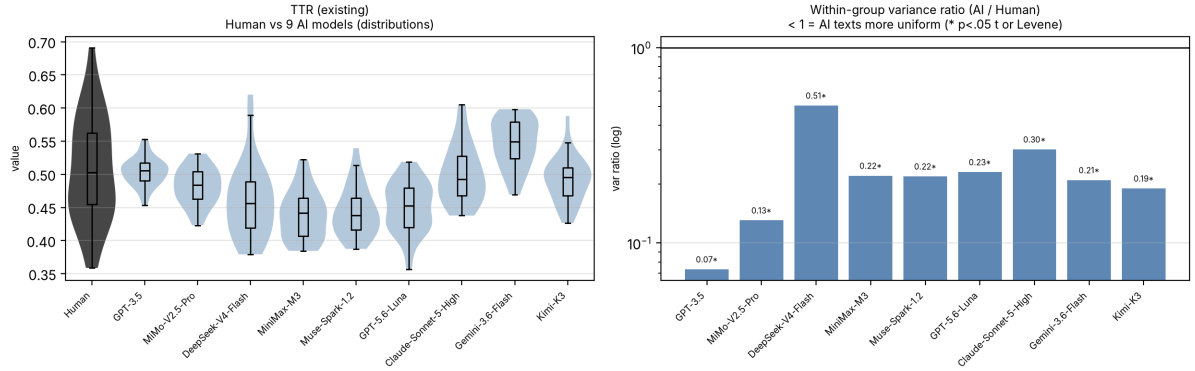


Figure 11: Figure 11 | Type-token ratio (TTR).

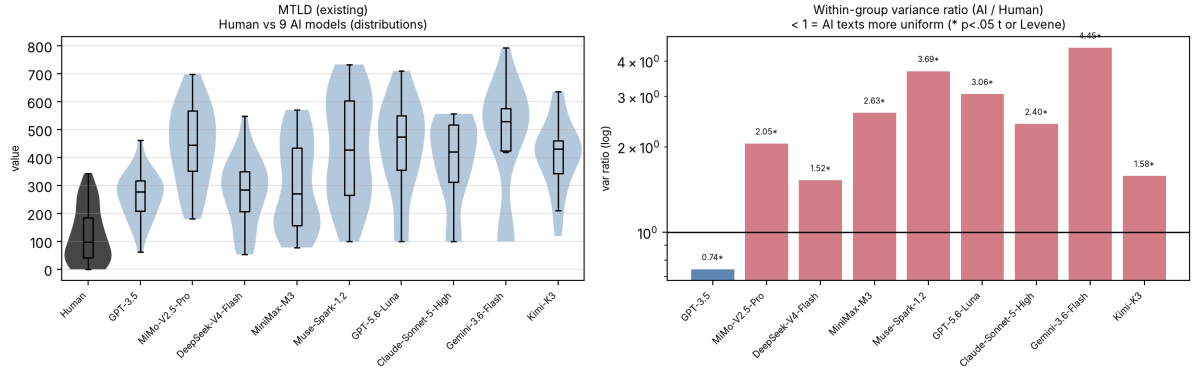


Figure 12: Figure 12 | MTLD.

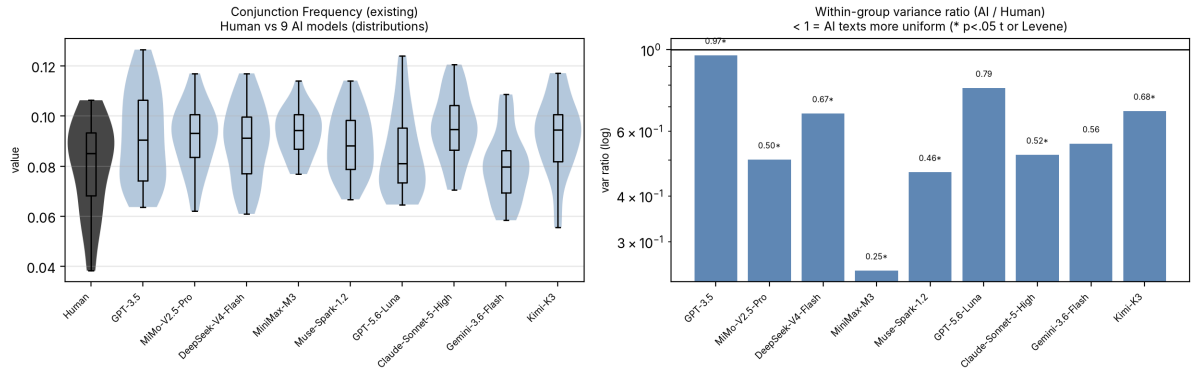


Figure 13: Figure 13 | Conjunction frequency.

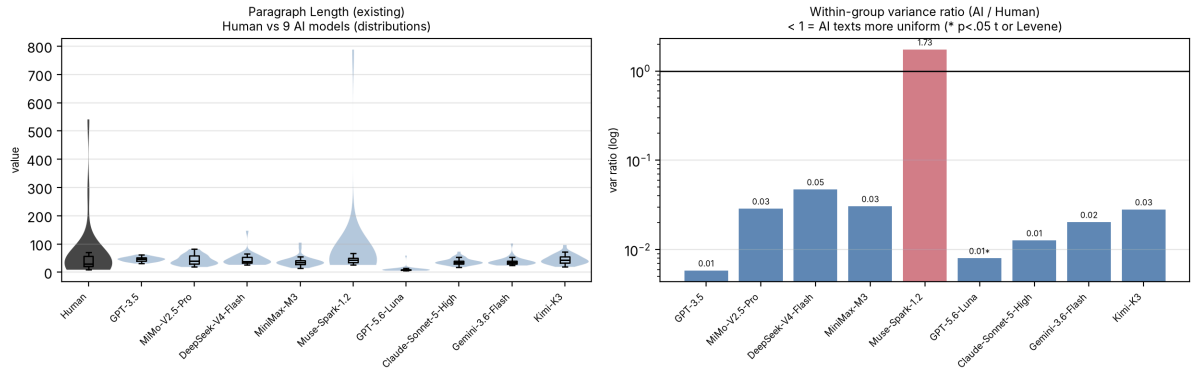


Figure 14: Figure 14 | Paragraph length.

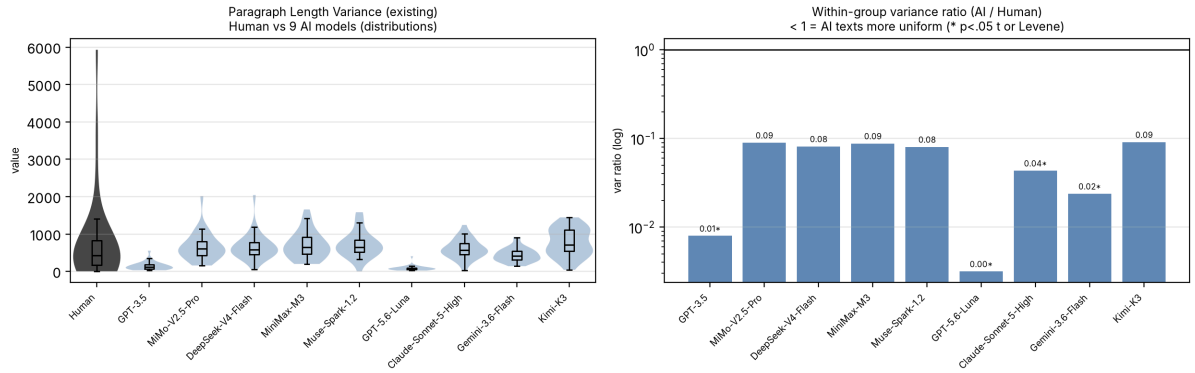


Figure 15: Figure 15 | Paragraph-length variance.

Appendix: BibTeX

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  title={Hierarchical Neural Story Generation},
  author={Fan, Angela and Lewis, Mike and Dauphin, Yann},
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